(Please write your Enrollment Number) Enrollment No. 0160117021

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:< B.Tech CSE AI/ ECE-AI >, Semester:<5th >
(November-December, 2023)

Subject Code: BAI 301	Subject: Machine Learning
Time : 3 Hours	Maximum Marks : 60
Note: Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.	

Q1.		(2.5*8=20)
	(a) With reference to logistic regression, if you will travel along with ROC curve, how cutoff value will change? Explain?	
	(b) A classifier is designed to identify if patients of a clinic need to go through the rest of the diagnostic steps after the first round of testing. What classification metric would be more or less appropriate? Why?	
	(c) Is it a good idea to focus on hyperparameter optimization when we can also improve the quality or quantity of the training data?	
	(d) A decision tree classifier learned from a fixed training set achieves 100% accuracy on the test set. Which of the algorithms (Logistic regression, An SVM with a polynomial kernel, k-Nearest neighbors, Naïve Bayesclassifier) trained using the same training set is guaranteed to give a model with 100% accuracy? Explain.	
	(e) What is the curse of dimensionality?	
	(f) If a Decision Tree is underfitting the training set, is it a good idea to try scaling the input features? Explain?	
	(g) A correlation between age and health of a person found to be 0.99 . On the basis of this you would tell the doctors that age causes health? Explain with reason?	
	(h) Suppose you are using Ridge Regression and you notice that the training error and the validation error are almost equal and fairly high. Would you say that the model suffers from high bias or high variance? Should you increase the regularization hyperparameter α or reduce it? Explain?	
	UNIT-1	
Q2.	(a) Discuss the potential consequences of not handling outliers during data preprocessing (b) How can feature engineering contribute to improving the performance of machine learning models, and what are some common techniques used in feature engineering? (c) How can you use quantiles of a distribution to detect its outliers?	(3+3+4=10)
Q3.	(a) What are different steps in data pre-processing required before any model building? Why these steps are required? (b) Explain the difference between the one-hot and label encoding methods? (c) Discuss the trade-offs between removing outliers and transforming them during the data preprocessing stage	(3+3+4=10)
	UNIT-2	
Q4.	A dataset of patients who have tested positive or negative for COVID-19 follows. Their symptoms are cough (C), fever (F), difficulty breathing (B), and tiredness (T).	(10)

	Cough (C)	Fever (F)	Difficulty breathing (B)	Tiredness (T)	Diagnosis
Patient 1	no	yes	yes	yes	Sick
Patient 2	yes	yes	no	yes	Sick
Patient 3	yes	no	yes	yes	Sick
Patient 4	yes	yes	yes	no	Sick
Patient 5	yes	no	no	yes	Healthy
Patient 6	no	yes	yes	no	Healthy
Patient 7	no	yes	no	no	Healthy
Patient 8	no	no	no	yes	Healthy

Build a naive Bayes model that predicts the diagnosis from the symptoms. Use the naive Bayes algorithm to find the following probabilities:

Note: For the following questions, the symptoms that are NO mentioned are completely unknown to us. For example, if we know that the patient has a cough, but nothing is said about their fever, it does not mean the patient doesn't have a fever.

a. The probability that a patient is sick given that the patient has a cough

b. The probability that a patient is sick given that the patient is not tired

c. The probability that a patient is sick given that the patient has a cough and a fever

d. The probability that a patient is sick given that the patient has a cough and a fever, but no difficulty breathing.

Q5. (a)If it takes one hour to train a Decision Tree on a training set containing 1 million instances, roughly how much time will it take to train another Decision Tree on a training set containing 10 million instances? (4+6 =10)

(b) Consider a dataset of labeled points in a 2D space:
Dataset:
Point 1: (2, 3), Label: Class A
Point 2: (5, 8), Label: Class B
Point 3: (1, 2), Label: Class A
Point 4: (8, 8), Label: Class B
Point 5: (7, 3), Label: Class B
Point 6: (6, 4), Label: Class B
Point 7: (1, 6), Label: Class A
Point 8: (4, 7), Label: Class B
Now, suppose you have a new unlabeled point: (3, 5).
Apply the k-Nearest Neighbors (KNN) algorithm to classify the new point. Set k=3 for this problem.
Provide the classification result using Euclidean distance for the new point and discuss any considerations or insights you might have regarding the choice of k and the impact of different distance metrics.

UNIT-3

Q6.	Given the following set of points: [(2, 3), (5, 8), (1, 2), (8, 8), (7, 3), (6, 4), (1, 6), (4, 7)] Compute full iterations of k-means clustering for K=2 with initial clusters Centroid $\mu1 = (1, 2)$ and $\mu2 = (6, 4)$. Make sure to write down the necessary distances (Euclidean distance), explain the steps you follow, and to describe the resulting clusters (centroid and points) at the end of each iterations.	(10)
Q7.	(a) What are the main motivations for reducing a dataset's dimensionality? What are the main drawbacks? (b) Can PCA be used to reduce the dimensionality of a highly nonlinear dataset? Explain?	(6+4 =10)

UNIT-4

Q8.	(a)A boosted strong learner L is formed by three weak learners, L1 , L2 , and L3 . Their weights are 1, 0.4, and 1.2, respectively. For a particular point, L1 and L2 predict that its label is positive, and L3 predicts that it's negative. What is the final prediction the learner L makes on this point? (b) What is the difference between hard and soft voting classifiers? (c) If your AdaBoost ensemble underfits the training data, which hyperparameters should you tweak and how?	(4+2+4=1 0)
Q9.	(a)If you have trained five different models on the exact same training data, and they all achieve 95% precision, is there any chance that you can combine these models to get better results? if so, how? If not, why? (b) Describe Random Forest in the context of ensemble learning. How does Random Forest utilize the concept of ensemble learning, and what distinguishes it from a single Decision Tree? Discuss the benefits of employing an ensemble of trees in Random Forest?	(5+5=10)